

Probabilistic Palmprint Embedding with Generative Regularization for Open-Set Identification

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Abstract

Open-set palmprint identification must identify enrolled subjects from imperfect images and reject subjects not in the gallery. The ROI of each palmprint is modeled as one diagonal Gaussian distribution. Its mean acts as the identity representation, and its variance provides an input-dependent estimate of aleatoric uncertainty [1, 2]. During training, an auxiliary U-Net decoder is conditioned on a sample from this distribution. This branch encourages the latent representation to retain information about palm-line structure, and is dropped at inference time [3, 4]. The training loss is composed of angular-margin classification, covariance regularization, image reconstruction and heteroscedastic uncertainty calibration. For open-set decisions, we consider similarity, probe uncertainty and symmetric KL divergence together. All decision thresholds are chosen on a different validation partition [5]. We evaluate four inference modes on seven backbone-dataset combinations. Direct matching in the latent-mean space leads to an improvement of 13.13 percentage points in the average open-set Rank-1 and a reduction of 11.45 points in the average EER over projected-mean matching. This trend holds in six settings, with IITD with ResNet18 as the exception. Thus, the experiments suggest that latent-mean matching is the most effective of the inference options we tested, but they do not clearly separate the contribution of each training element.

Keywords: Palmprint recognition, probabilistic embedding, open-set identification, aleatoric uncertainty, generative regularization

1 Introduction

Palmprint recognition is based on principal lines, wrinkles and ridge patterns found on a relatively large biometric surface. Modern convolutional models achieve good performance under controlled acquisition [gaoDeepLearningPalmprint2025,yangComprehensiveCompetitionMechanism2023]. Touchless and mobile capture are less predictable. Illumination, hand pose, motion blur, sensor response, and non-rigid skin deformation can alter the visible crease pattern and lead to poorer consistency of features extracted from repeated samples [4, 6, 7].

In most recognition systems an image is mapped to a single deterministic feature vector. This representation is sufficient for ordinary matching but it does not tell how reliable the input was. Even a palmprint that is blurred or in the wrong place can give a confident similarity score. This is particularly an issue in open-set identification, where the system must identify enrolled identities while rejecting probes from identities that have never been enrolled [5].

We solve this problem by predicting a Gaussian embedding for each palmprint. The mean includes the identity information used by the classifier, while the diagonal variance describes the sample-dependent uncertainty [1, 2]. We also attach a reconstruction decoder in training. This decoder is conditioned on a stochastic latent sample, while the recognition branch is based on the deterministic mean. The reconstruction task regularizes the palm-line information, and is not used as evidence of topological preservation nor does it add any inference-time cost.

The main contributions are:

- We model palmprint features as diagonal Gaussian embeddings and explicitly decouple the mean, variance, stochastic sample, projection head and reconstruction decoder.
- We train the representation with an identity, covariance, reconstruction and uncertainty-calibration objective under a staged schedule.
- We define four inference modes and a validation-calibrated rejection rule combining similarity, probe uncertainty and symmetric distribution distance.
- We report results for seven backbone-dataset combinations, including the IITD-ResNet18 case where latent-mean matching does not beat projected-mean matching.

2 Related Work

2.1 Palmprint Representation Learning

Previous palmprint systems used hand-crafted line and texture descriptors. Recent work has moved to learned convolutional features. PalmNet uses Gabor processing, principal-component analysis and convolutional learning for touchless recognition [6]. CCNet employs competition mechanisms to improve class separability [8]. However,

cross-dataset studies are still very sensitive to acquisition devices, ROI extraction and hand-pose distributions [7]. Usually these methods produce point embeddings and the quality of the image is reflected indirectly in the location of the feature.

Our identity loss is an angular-margin classification objective. It is different from pairwise or instance-level contrastive learning, although both families aim to improve separation in the feature space [9, 10]. To avoid conflating these objectives we will use the term *angular-margin classification* throughout the paper.

2.2 Probabilistic Embeddings and Uncertainty

Probabilistic Face Embeddings (PFE) [1] represent each image as a Gaussian distribution, incorporating both the mean and variance during the matching process. Data Uncertainty Learning also learns input-dependent variance so that low-quality samples are treated with less confidence [2]. Deep Variational Metric Learning constrains the metric learning variationally and models the intra-class variation within a latent distribution [11]. These studies motivate our Gaussian feature head. However, in the palmprint setting, the identity evidence is strongly associated with local line and texture patterns. This motivates the additional reconstruction branch.

The variance of the model that is predicted is the aleatoric uncertainty that is dependent on the image observed. It does not model uncertainty in the parameters of the network. Thus epistemic uncertainty is out of scope of the present model.

2.3 Open-Set Recognition

Open-set identification combines gallery matching with a rejection decision. There are high but unreliable similarity scores for unknown identities, low quality probes and crowded gallery neighborhoods. Gallery-aware uncertainty methods combine the probe’s information with ambiguity about the enrolled set [5]. Our decision rule is of this general idea, but all thresholds are selected on validation data and then fixed prior to test evaluation.

2.4 Generative Regularization

An encoder-decoder objective might constrain the latent representation to contain information that will be needed to reconstruct the input. The Probabilistic U-Net is a conditional latent-variable model for ambiguous image outputs, while preserving spatial information through a U-Net decoder [3]. A related approach to separate salient and background variation is contrastive variational autoencoders [12]. Diff-Palm also shows that palmprint synthesis needs to consider crease geometry and non-rigid intra-class change [4]. In our model we use only the decoder as a training regularizer. It does not produce identities for recognition, nor does it perform inference.

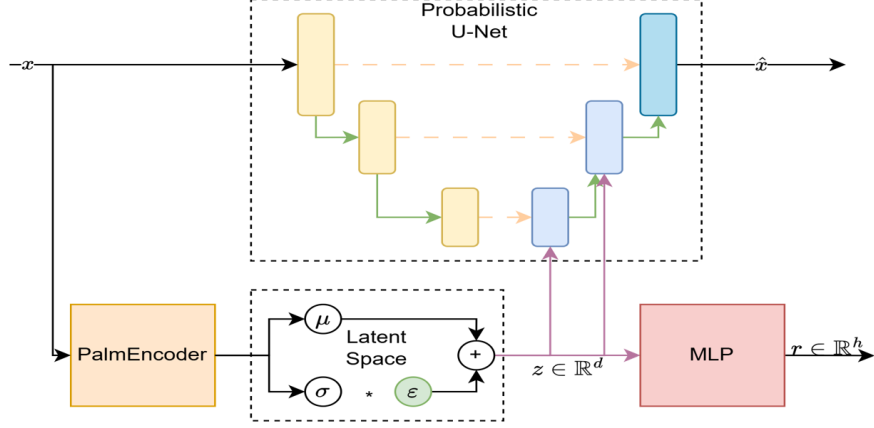


Fig. 1: Overall framework. The PalmEncoder predicts the Gaussian parameters (μ, σ^2) . The mean μ is projected by the MLP for identity learning, whereas a stochastic sample $\mathbf{z}$ conditions the U-Net decoder during training. The reconstruction branch is removed at inference.

3 Method

3.1 Architecture and Notation

For sample i , $\mathbf{x}_i \in \mathbb{R}^{C \times H \times W}$ denotes the normalized palmprint ROI and y_i its identity label. The encoder E_θ produces a shared feature vector

$$\mathbf{h}_i = E_\theta(\mathbf{x}_i) \in \mathbb{R}^{d_h}. \quad (1)$$

Two linear heads are applied to $\mathbf{h}_i$. The first returns the identity mean $\mu_i \in \mathbb{R}^d$. The second returns $\rho_i \in \mathbb{R}^d$, which is converted to a positive variance for each latent dimension:

$$\sigma_i^2 = \text{softplus}(\rho_i) + \varepsilon_\sigma, \quad \varepsilon_\sigma > 0, \quad (2)$$

and the probabilistic embedding is

$$q_\theta(\mathbf{z}_i | \mathbf{x}_i) = \mathcal{N}(\mu_i, \text{diag}(\sigma_i^2)). \quad (3)$$

Equation (3) uses one variance value per latent dimension rather than a shared scalar. We sample the latent code by reparameterization:

$$\mathbf{z}_i = \mu_i + \sigma_i \odot \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (4)$$

The recognition branch maps the mean through an MLP, $\mathbf{r}_i = P_\psi(\mu_i) \in \mathbb{R}^h$. The decoder receives the sampled code $\mathbf{z}_i$ instead. Thus, identity supervision is deterministic for a given input, while the reconstruction branch remains stochastic.

Figure 1 shows the two routes. The projected mean is used for identity classification. In the auxiliary route, encoder skip features provide local detail, and $\mathbf{z}_i$ modulates

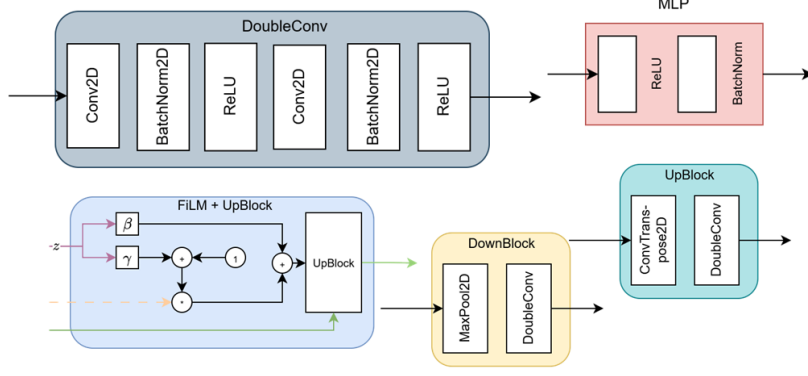


Fig. 2: Internal blocks used by the encoder-decoder. The FiLM parameters (γ, β) are predicted from $\mathbf{z}$ and modulate decoder features before the upsampling block.

the decoder blocks through feature-wise linear modulation [13]. The decoder output is $\hat{\mathbf{x}}_i$.

3.2 Identity and Variance Objectives

The classifier receives the normalized projection of the mean

$$\bar{\mathbf{r}}_i = \frac{P_\psi(\boldsymbol{\mu}_i)}{\|P_\psi(\boldsymbol{\mu}_i)\|_2}. \quad (5)$$

With normalized class weights $\mathbf{w}_j$, angular margin m , and scale s , the additive angular-margin loss is

$$\mathcal{L}_{\text{id}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp[s \cos(\theta_{y_i} + m)]}{\exp[s \cos(\theta_{y_i} + m)] + \sum_{j \neq y_i} \exp[s \cos(\theta_j)]}. \quad (6)$$

Using $\boldsymbol{\mu}_i$ rather than $\mathbf{z}_i$ prevents the class score from depending on a particular random sample.

A KL penalty against $\mathcal{N}(\mathbf{0}, \mathbf{I})$ would regularize both the mean and covariance. We regularize only the covariance because pulling identity means toward the origin can conflict with class separation. This is equivalent to matching $\mathcal{N}(\boldsymbol{\mu}_i, \text{diag}(\boldsymbol{\sigma}_i^2))$ to $\mathcal{N}(\boldsymbol{\mu}_i, \mathbf{I})$:

$$\mathcal{L}_{\text{var}} = \frac{1}{2Nd} \sum_{i=1}^N \sum_{k=1}^d (\sigma_{i,k}^2 - \log \sigma_{i,k}^2 - 1). \quad (7)$$

This penalty grows when a variance collapses toward zero or becomes excessively large, but it does not move the identity mean.

3.3 Generative and Uncertainty Objectives

The reconstruction loss combines an ℓ_1 pixel term with an ℓ_1 penalty on first-order image gradients:

$$\mathcal{L}_{\text{rec}} = \frac{1}{N} \sum_{i=1}^N [\|\mathbf{x}_i - \hat{\mathbf{x}}_i\|_1 + \lambda_{\nabla} \|\nabla \mathbf{x}_i - \nabla \hat{\mathbf{x}}_i\|_1]. \quad (8)$$

The gradient term emphasizes line boundaries and crease transitions. It should be interpreted as an edge-sensitive reconstruction term, not as a guarantee of topological invariance.

For sample i , we summarize reconstruction error by e_i and predicted uncertainty by the mean log-variance u_i :

$$e_i = \frac{1}{CHW} \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|_1, \quad u_i = \frac{1}{d} \sum_{k=1}^d \log \sigma_{i,k}^2. \quad (9)$$

We use a heteroscedastic objective [2]

$$\mathcal{L}_{\text{unc}} = \frac{1}{N} \sum_{i=1}^N \left(\frac{1}{2} \exp(-u_i) e_i + \frac{1}{2} u_i \right), \quad (10)$$

The first term increases the loss when a large residual is paired with low predicted variance; the second prevents the model from reducing that penalty by increasing variance without bound.

The total objective is

$$\mathcal{L}(t) = \mathcal{L}_{\text{id}} + \lambda_{\text{var}}(t) \mathcal{L}_{\text{var}} + \lambda_{\text{rec}} \mathcal{L}_{\text{rec}} + \lambda_{\text{unc}} \mathcal{L}_{\text{unc}}. \quad (11)$$

We first optimize the identity objective for T_{w} epochs. The reconstruction and uncertainty terms are then enabled, while

$$\lambda_{\text{var}}(t) = \lambda_{\text{var}}^{\max} \min \left(1, \frac{\max(0, t - T_{\text{w}})}{T_{\text{a}}} \right) \quad (12)$$

increases the variance weight linearly over T_{a} epochs. This ordering gives the identity branch an initial period of optimization before covariance and reconstruction terms are introduced.

3.4 Open-Set Inference

At inference time, only the encoder and embedding heads are retained. For a probe p and gallery template g , latent-mean similarity is computed as

$$S_{\mu}(p, g) = \frac{\boldsymbol{\mu}_p^{\top} \boldsymbol{\mu}_g}{\|\boldsymbol{\mu}_p\|_2 \|\boldsymbol{\mu}_g\|_2}. \quad (13)$$

We summarize probe uncertainty by the arithmetic mean of its predicted variances:

$$U_p = \frac{1}{d} \sum_{k=1}^d \sigma_{p,k}^2. \quad (14)$$

We also measure distributional compatibility using the symmetric KL divergence

$$D_{\text{SKL}}(p, g) = \frac{1}{2} [D_{\text{KL}}(q_p \| q_g) + D_{\text{KL}}(q_g \| q_p)], \quad (15)$$

where, for diagonal Gaussians,

$$D_{\text{KL}}(q_p \| q_g) = \frac{1}{2} \sum_{k=1}^d \left[\frac{\sigma_{p,k}^2 + (\mu_{p,k} - \mu_{g,k})^2}{\sigma_{g,k}^2} - 1 + \log \frac{\sigma_{g,k}^2}{\sigma_{p,k}^2} \right]. \quad (16)$$

Let $g^* = \arg \max_g S(p, g)$ under the selected inference mode. The probe is accepted as identity g^* only if

$$U_p \leq \tau_U, \quad S(p, g^*) \geq \tau_S, \quad D_{\text{SKL}}(p, g^*) \leq \tau_K. \quad (17)$$

The threshold triplet (τ_U, τ_S, τ_K) is selected jointly on the validation split and then kept fixed. Unknown identities used for validation do not appear in training set or in either test partition. No test sample is used to select an EER or rejection operating point.

Inference modes.

Table 1 compares four scoring procedures. We first define

$$S_{\text{proj}}(p, g) = \frac{P_\psi(\boldsymbol{\mu}_p)^\top P_\psi(\boldsymbol{\mu}_g)}{\|P_\psi(\boldsymbol{\mu}_p)\|_2 \|P_\psi(\boldsymbol{\mu}_g)\|_2}. \quad (18)$$

- **M0: projected-mean matching.** All gallery templates are scored with $S_{\text{proj}}(p, g)$.
- **M1: projected-space adaptation.** First, $g_0 = \arg \max_g S_{\text{proj}}(p, g)$ is selected. A residual $\boldsymbol{\delta}_r$ is then optimized for a fixed number of steps:

$$\boldsymbol{\delta}_r^* = \arg \min_{\boldsymbol{\delta}_r} [-\cos(\mathbf{r}_p + \boldsymbol{\delta}_r, \mathbf{r}_{g_0}) + \eta_r \|\boldsymbol{\delta}_r\|_2^2]. \quad (19)$$

The candidate remains g_0 , and its adapted score is

$$S_{\text{M1}}(p, g_0) = \cos(\mathbf{r}_p + \boldsymbol{\delta}_r^*, \mathbf{r}_{g_0}). \quad (20)$$

- **M2: latent-space adaptation.** Starting from the same g_0 , a residual $\boldsymbol{\delta}_\mu$ is optimized before projection:

$$\boldsymbol{\delta}_\mu^* = \arg \min_{\boldsymbol{\delta}_\mu} [D_{\text{SKL}}(\mathcal{N}(\boldsymbol{\mu}_p + \boldsymbol{\delta}_\mu, \text{diag}(\boldsymbol{\sigma}_p^2)), q_{g_0}) + \eta_\mu \|\boldsymbol{\delta}_\mu\|_2^2]. \quad (21)$$

The candidate remains g_0 , and its adapted similarity score is

$$S_{M2}(p, g_0) = \cos(P_\psi(\boldsymbol{\mu}_p + \boldsymbol{\delta}_\mu^*), P_\psi(\boldsymbol{\mu}_{g_0})). \quad (22)$$

- **M3: latent-mean matching.** All gallery templates are scored with $S_\mu(p, g)$, without test-time optimization.

For M0 and M3, g^* in Eq. (17) is obtained by scoring the full gallery. For M1 and M2, the candidate remains the initially selected g_0 . Neither adaptation method uses the probe label. The number of steps, step size, and residual penalties are chosen on validation data and reused unchanged for the test set.

The source code is publicly available at [our GitHub repository](#).

4 Experiments

4.1 Protocol and Reproducibility

Three datasets were used, which include the authors’ Own, Tongji and IITD. The dataset is split into training, validation, enrolled-test, and unknown-test sets. The enrolled-test identities supply both the gallery and the known probes. The identities in the unknown-test partition are not used for training, validation and enrollment. For identities with multiple enrollment images, we compute a gallery template by averaging the predicted means and diagonal variances.

CCNet [8] and ResNet18 are evaluated on all three datasets. PalmNet [6] is evaluated on the Own dataset only, giving seven backbone-dataset combinations.

All experiments in a dataset are processed with the same ROI extraction and normalization procedure. Training augmentation includes illumination changes, blur, noise and moderate geometric deformation. The exact ranges have to be the same as the ones in the training configuration. We use stochastic gradient descent with momentum, an identity-only warm-up, and scheduled learning-rate decay to optimize the models.

The validation procedure for selecting (τ_U, τ_S, τ_K) is an automated grid search over the validation split to determine the optimal operating point that maximizes recognition accuracy while minimizing false acceptances.

We report Rank-1 accuracy for identification and EER, FAR and FRR for verification and rejection. S1-FRR is the false-rejection rate due to the uncertainty condition of Eq. (17). FAR, FRR and S1-FRR are evaluated at thresholds set on the validation split. The open-set Rank-1 is defined as the percentage of enrolled (known) probes that are correctly matched to their true identity at Rank-1 and successfully pass all rejection thresholds. Any known probe that is rejected is counted as an identification failure. Unknown probes are not included in this denominator and are evaluated separately via the False Acceptance Rate (FAR).

Table 1: Reported performance (%) for four inference modes applied to the trained probabilistic model. Bold values are the best within each dataset-backbone block; Rank-1 is maximized, and all error rates are minimized.

Dataset	Backbone	Mode	Closed-set		Open-set		Verification		
			Rank-1	EER	Rank-1	EER	FAR	FRR	S1-FRR
Own	CCNet	M0	33.88	25.27	34.19	59.75	52.87	66.43	53.51
		M1	92.05	8.55	92.05	20.71	27.22	56.07	51.68
		M2	87.97	10.59	88.99	22.60	25.65	56.78	51.68
		M3	94.60	3.39	94.60	11.33	33.86	53.21	51.68
	ResNet18	M0	80.84	7.03	79.20	24.03	41.59	53.41	50.15
		M1	81.14	6.91	79.61	27.42	45.90	52.70	50.15
		M2	67.07	12.83	74.72	33.00	39.95	56.47	50.15
		M3	84.20	5.82	82.57	21.05	36.66	52.50	50.15
	PalmNet	M0	78.90	8.89	83.38	23.50	30.57	49.13	44.34
		M1	81.24	6.82	82.87	31.50	28.80	48.01	44.34
		M2	76.96	9.69	81.75	33.92	30.37	49.54	44.34
		M3	92.46	3.66	92.56	13.99	31.60	46.18	44.34
Tongji	CCNet	M0	40.95	18.70	39.66	59.24	44.06	56.90	45.69
		M1	38.79	25.09	38.79	61.33	41.45	60.34	45.69
		M2	34.48	28.17	36.21	60.68	40.29	61.21	45.69
		M3	55.17	12.95	54.31	50.94	44.06	51.72	45.69
	ResNet18	M0	83.60	6.40	83.60	25.20	42.20	51.60	46.40
		M1	79.00	6.57	79.00	33.75	42.20	51.00	46.40
		M2	72.20	8.42	72.20	36.60	42.10	52.00	46.40
		M3	86.80	5.23	86.80	21.55	38.10	50.60	46.40
IITD	CCNet	M0	95.20	2.70	96.60	13.25	17.80	48.80	46.40
		M1	96.40	4.15	97.00	18.45	16.90	48.40	46.40
		M2	88.80	6.55	92.20	25.50	16.60	49.00	46.40
		M3	99.40	0.42	99.40	4.55	6.60	46.60	46.40
	ResNet18	M0	96.67	1.99	96.67	7.46	28.16	58.33	56.67
		M1	95.83	6.11	95.83	16.67	24.71	60.00	56.67
		M2	92.50	7.20	92.50	21.62	25.29	61.67	56.67
		M3	95.00	3.58	95.00	8.89	34.48	58.33	56.67

4.2 Quantitative Results

M3 achieves the best closed-set Rank-1 and EER on six out of the seven backbone-dataset pairs. The same six settings also prefer M3 for open-set Rank-1 and EER. On average across all combinations, moving from M0 to M3 increases the open-set Rank-1 by 13.13 percentage points and decreases the open-set EER by 11.45 points. The most prominent change is in Own-CCNet where the open-set Rank-1 increases from 34.19% to 94.60%, and the EER decreases from 59.75% to 11.33%.

M1 and M2 do not exhibit the same level of consistency. Both approaches start with the gallery candidate selected in the projected space and keep it during adaptation. Thus, their worse performance is consistent with a failure mode in which an incorrect initial candidate is reinforced rather than corrected. This interpretation remains a hypothesis until validated by candidate level error analysis. The most obvious exception to the overall M3 trend is IITD-ResNet18: M0 achieves 96.67% open-set Rank-1

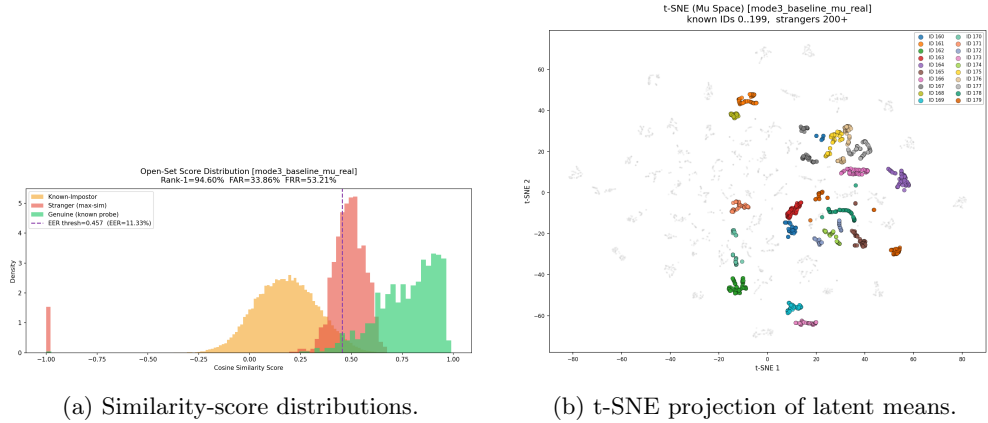


Fig. 3: Visual analysis for the TBD-CCNet experiment. Panel (a) compares genuine, known-impostor, and unknown-probe scores. Panel (b) projects the latent means of enrolled and unseen identities with t-SNE.

and 7.46% EER, while M3 achieves 95.00% and 8.89%, respectively. M1 has the lowest FAR among this block, but higher EER and FRR.

The table also highlights that identification and rejection metrics are not always in favor of the same mode. M2 has the lowest FAR for Own-CCNet and Tongji-CCNet, while M3 has higher identification accuracy for both. Therefore, for deployment, the operating point should be selected from the target FAR or FPIR requirement, not from EER alone.

4.3 Distribution Visualization

The overlap of genuine, known-impostor and unknown-probe scores is shown directly in Figure 3a. Figure 3b shows the two-dimensional t-SNE projection of local clusters of selected latent means. These plots are only descriptive: t-SNE does not preserve global distances and a single score histogram cannot establish calibration. Quantitative open-set analysis should also report DIR or TPIR at fixed FPIR, OSCR, known-versus-unknown AUROC, and risk-coverage curves if available.

5 Discussion and Limitations

The most consistent result in Table 1 is that direct latent mean matching is better than either adaptation procedure. Methods M1 and M2 optimize a probe for the candidate selected before adaptation, and no method searches the gallery again after that. An incorrect initial candidate can thus remain locked in. A candidate-level analysis is still needed to determine how often this mechanism accounts for the observed errors. The weaker M3 result on IITD-ResNet18 also suggests that the preferred matching space depends on the backbone and dataset.

The present experiments compare paths of inference within a single probabilistically trained model. They do not compare the entire approach to a deterministically

trained independent model. Thus, the results warrant a conclusion regarding representation choice-the latent mean is generally stronger than the projected mean in this implementation-but do not quantify the separate contributions of the Gaussian head, reconstruction decoder, variance regularizer, uncertainty loss or individual rejection conditions. These contributions need controlled ablations and repeated runs with uncertainty intervals.

Several parts of the evaluation need to be done to be able to reproduce the results. For the Own dataset, a documented acquisition procedure, split statistics, consent or ethics information where applicable, and a data-access statement are required. It will also report the ROI parameters, augmentation ranges, the software versions, hardware and all the training hyperparameters. The quality of the reconstruction has not been yet explored using image examples or a crease sensitive structural measure. Also, the present evaluation does not consider DIR or TPIR at fixed FPIR, OSCR, calibration curves or risk-coverage analysis. Finally, the predicted variance only accounts for aleatoric uncertainty; we do not consider parameter uncertainty, adversarial manipulation, or cross-domain shift.

6 Conclusion

We proposed a palmprint representation where each ROI is modeled by a diagonal Gaussian. The mean is used by the angular-margin identity branch, while a sampled latent vector conditions the auxiliary reconstruction decoder during training. The decoder is dropped at inference time. We accept a probe only if its similarity, predicted uncertainty and symmetric Kullback-Leibler distance meet thresholds chosen on validation data.

Among the four inference modes we evaluate, latent-mean matching achieves the best open-set Rank-1 and EER in six of seven backbone-dataset settings. The only exception is IITD-ResNet18, where projected-mean matching works better. The current results indicate latent-mean matching as the strongest inference option in this implementation. However, the results are not sufficient to assign the gain to probabilistic training or generative regularization. That question requires deterministic baselines, component-wise ablations, full protocol reporting and repeated experiments.

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